

The Islamia University of Bahawalpur

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SRS REPORT

for

AI Interview Practice and Feedback System

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1. Introduction

The AI-Based Interview Practice and Feedback System is a web-based application developed as a Final Year Project at the Islamia University of Bahawalpur. The system provides an interactive platform where students and job seekers can practice interview questions related to different job roles. It uses Natural Language Processing (NLP) techniques to analyze user responses and provide feedback and scoring. The system aims to improve interview preparation by offering a smart and automated solution that helps users identify their weaknesses and improve their communication and technical responses.

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements of the AI-Based Interview Practice and Feedback System.

This document serves as a reference for developers, supervisors, and evaluators by clearly describing:

- what the system will do
- how it will behave
- what features it will include

The main objective of this system is to provide an intelligent platform that assists users in preparing for job interviews through automated evaluation and feedback.

1.2 Scope

The AI Interview Practice System will be developed as a web-based application that allows users to:

- register and log into the system
- select job roles (e.g., software developer, data analyst)
- answer interview questions

- receive automated feedback and scores

The system will include:

- a user interface for interaction
- a backend system for processing
- an NLP module for analyzing answers

Out of Scope:

- mobile application development
- real-time video interview simulation
- advanced speech recognition
-

1.3 Product Perspective

The system follows a client-server architecture.

- The frontend is developed using HTML, CSS, and JavaScript to provide an interactive user interface.
- The backend is implemented using Python and Flask to handle system logic and user requests.
- The NLP module uses NLTK to analyze user answers and generate feedback.
- The database (SQLite/MySQL) stores user data, questions, and responses.

The system works as a standalone web application and can be accessed through any modern web browser.

1.4 User Characteristics

The system includes the following types of users:

Students

- Primary users of the system
- Have basic knowledge of web applications
- Use the system to practice interview questions

Job Seekers

- Individuals preparing for job interviews
- Use the system to improve their responses and confidence

Administrator

- Manages interview questions and system data
- Has moderate technical knowledge
- Responsible for maintaining the system

1.5 Proposed Technologies

The system will be developed using the following technologies:

Category	Technology	Purpose
Frontend	HTML, CSS, JavaScript	User interface
Backend	Python (Flask)	Server-side logic
NLP	NLTK	Text analysis
Database	SQLite / MySQL	Data storage
Version Control	GitHub	Code management

2. Requirements

The AI-Based Interview Practice and Feedback System is a web-based application that allows users to practice interview questions and receive automated feedback. The system performs core operations such as user registration, login, job selection, question display, answer submission, NLP-based analysis, and feedback generation.

The system ensures a simple and interactive experience for users by combining web technologies with Natural Language Processing techniques.

2.1.1 User Registration (Sign Up)

Name	FR001
Purpose	Allow a new user to create an account on the system to access interview practice features.
User(s)	Student, Job Seeker
Input	• Full Name: Name of the user • Email Address: Unique email • Password: Minimum 8 characters • Role: User type
Output	User account created and stored in database. User redirected to login page.

2.1.2 User Login

Name	FR002
Purpose	Authenticate a registered user and allow access to system.
User(s)	Student, Job Seeker
Input	• Email Address • Password
Output	User successfully logs in and is redirected to dashboard.

2.1.3 Select Job Role

Name	FR003
Purpose	Allow user to select a job role for interview practice.
User(s)	Student, Job Seeker
Input	Job Title (e.g., Software Developer, Data Analyst)
Output	System loads interview questions based on selected job role.

2.1.4 Answer Questions

Name	FR004
Purpose	Allow users to submit answers to interview questions.
User(s)	Student, Job Seeker
Input	Text answers entered by user
Output	Answers are stored in the database.

2.1.5 Analyze Answers

Name	FR005
Purpose	Analyze user answers using NLP techniques.

Name	FR005
User(s)	System
Input	• User answers
Output	Keywords extracted and answers evaluated.

2.1.6 Feedback Generation

Name	FR006
Purpose	Provide feedback and scoring based on analysis.
User(s)	System
Input	• Analysis results
Output	Score and improvement suggestions displayed to user.

2.2 Non-Functional Requirements

The system must satisfy the following quality requirements:

Category	Requirement
Performance	System should respond quickly (within few seconds)

Category	Requirement
Security	User data must be protected
Usability	System should be easy to use
Availability	System should be available 24/7
Reliability	System should work without errors

3. Use Cases and Flow of Processes

The following use cases describe how users interact with the AI-Based Interview Practice and Feedback System. These use cases represent the main functionalities of the system and show how different components work together.

3.1 Use Case 1 – User Registration

ID	UC001
Name	User Registration
Description	This use case describes how a new user creates an account in the system by providing required information.
Requirement(s)	FR001
Actor(s)	Student, Job Seeker
Precondition	User must not already have an account.

ID	UC001
Postcondition	New user account is created and stored in database.
Basic Flow	1. User opens registration page. 2. User enters required details (name, email, password). 3. System validates input. 4. System saves user data in database. 5. User is redirected to login page.
Alternative Flow	Admin may create user account manually.
Exceptions	• Invalid email format • Weak password • Email already exists

3.2 Use Case 2 – Interview Practice

ID	UC002
Name	Interview Practice
Description	This use case describes how a user selects a job role and answers interview questions.
Requirement(s)	FR003, FR004
Actor(s)	Student, Job Seeker
Precondition	User must be logged in.

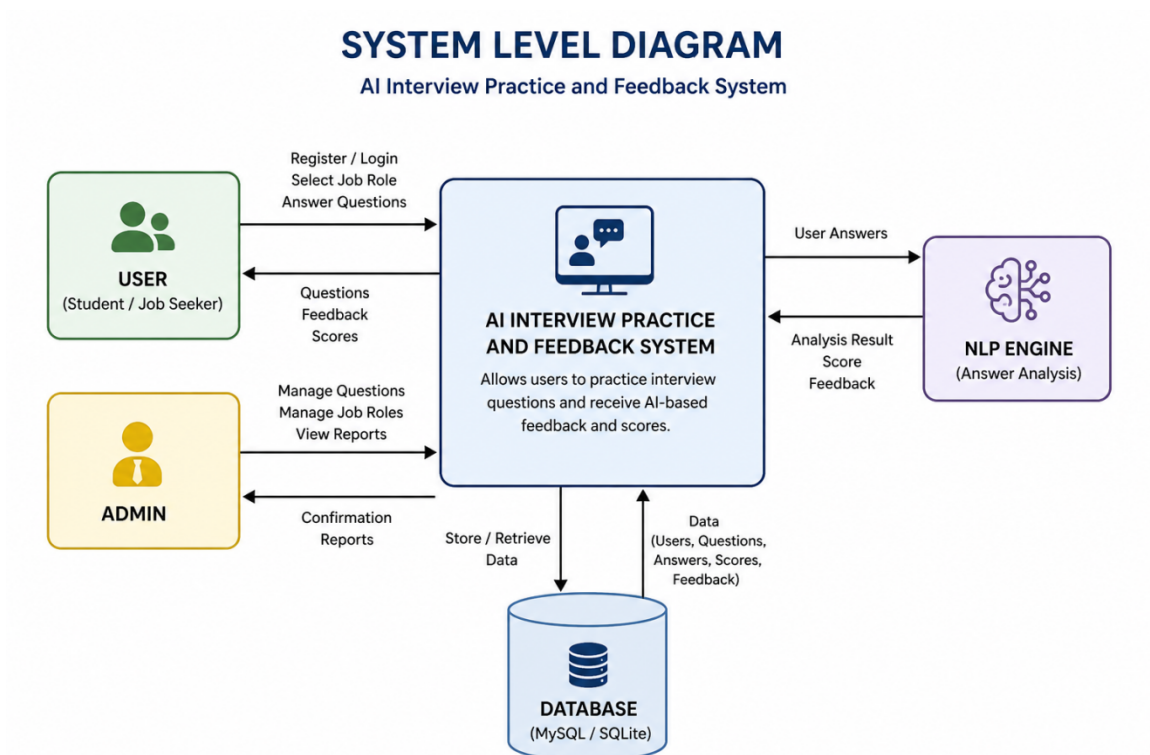
ID	UC002
Postcondition	User answers are stored in system.
Basic Flow	1. User logs into system. 2. User selects job role. 3. System displays interview questions. 4. User submits answers. 5. System stores answers.
Alternative Flow	User can change job role before submitting answers.
Exceptions	• User not logged in • No questions available

3.3 Use Case 3 – Feedback

ID	UC003
Name	Feedback Generation
Description	This use case describes how the system analyzes answers and provides feedback to the user.
Requirement(s)	FR005, FR006
Actor(s)	System
Precondition	User must submit answers.
Postcondition	User receives score and feedback.

ID	UC003
Basic Flow	1. System receives user answers. 2. NLP module analyzes answers. 3. System calculates score. 4. System generates feedback. 5. Feedback displayed to user.
Alternative Flow	User can retake interview to improve score.
Exceptions	• Invalid input • System processing error

4. System Architecture Diagram



5. References

The following resources were used in the development and understanding of this project:

1. Python Documentation – <https://docs.python.org>
2. Flask Documentation – <https://flask.palletsprojects.com>
3. NLTK Documentation – <https://www.nltk.org>
4. HTML, CSS, JavaScript Documentation –
<https://developer.mozilla.org>
5. GitHub Documentation – <https://docs.github.com>